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Refrigerating appliance with a fan and with a pressure sensor

Abstract

A refrigerating appliance (**100**) is provided, which comprises: —at least one storage compartment (**110**) for storing goods to be refrigerated; —a refrigeration circuit comprising at least one evaporator (**170**) associated with said at least one storage compartment (**110**); —a control unit (**172**) configured to control operation of the refrigerating appliance (**100**); —at least one fan (**171**), configured to promote heat exchange between said at least one evaporator (**170**) and the at least one storage compartment (**110**), said at least one fan (**171**) being further configured to be switched between a first operative condition in which said at least one fan (**171**) is commanded to rotate, and a second operative condition in which said at least one fan (**171**) is commanded to not rotate; —a MEMS pressure sensor (**180**) configured to measure the pressure inside said at least one storage compartment (**110**) and in signal communication with the control unit (**172**) for providing a pressure signal proportional to the measure of the pressure inside said at least one storage compartment (**110**), wherein the control unit (**172**) is configured to control the operation of the refrigerating appliance (**100**) in function of said pressure signal received when said at least one fan (**171**) is in the first operative condition.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8346398	12/2012	Ahmed	700/32	H05K 7/20836
9235219	12/2015	Arunasalam	N/A	G01F 1/69
9250008	12/2015	Kim	N/A	F25D 23/028
9605884	12/2016	Gomes	N/A	F25D 11/022
10808985	12/2019	Ji	N/A	F25D 29/00
2005/0258260	12/2004	Ahmed	N/A	N/A
2008/0072611	12/2007	Ahmed	62/132	F25B 49/02
2014/0150477	12/2013	Qu et al.	N/A	N/A
2019/0285331	12/2018	Kim	N/A	F25D 17/06

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3540343	12/2018	EP	N/A
WO-2021126958	12/2020	WO	G01D 11/00

OTHER PUBLICATIONS

Machine translation of WO2021126958 (Year: 2021). cited by examiner
International Search Report for PCT/EP2020/068421, dated Mar. 25, 2021, 2 pages. cited by applicant

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Background/Summary

FIELD OF THE INVENTION
(1) The present invention generally relates to a refrigerating appliance. More particularly, the present invention relates to a refrigerating appliance equipped with a pressure sensor system.
BACKGROUND OF THE INVENTION
(2) Among the available sensor devices, pressure sensors are sensor devices configured to measure the pressure of gases or liquids. These sensors can be expediently used in a wide range of different applications, such as for example for measuring the pressure inside of spaces (e.g., a room, a compartment, the environment), measuring altitude, measuring the flow of a fluid, and detecting fluid leaks.
(3) Modern refrigerating appliances having one or more storage compartments for refrigerating food and beverage articles can be equipped with one or more sensor devices, configured to measure the pressure inside said one or more storage compartments, and control the operation of the refrigerating appliance according to the measured pressure. For example, a pressure sensor may be used in order to assess a closed/open condition of a storage compartment door based on the measured pressure inside the refrigerating appliance.
SUMMARY OF INVENTION
(4) It is an object of the present invention to provide a refrigerating appliance equipped with a pressure sensor system that allows to manage and control in an efficient way the operation of the refrigerating appliance itself.
(5) More in detail, it is an object of the present invention to provide a refrigerating appliance having an improved temperature control, optimized power consumptions, and provided with an enhanced fault verify function.
(6) One or more aspects of the present invention are set out in the independent claims, with advantageous features of the same invention that are indicated in the dependent claims.
(7) An aspect of the present invention relates to a refrigerating appliance.
(8) According to an embodiment of the present invention, the refrigerating appliance comprises: at least one storage compartment for storing goods to be refrigerated; a refrigeration circuit comprising at least one evaporator associated with said at least one storage compartment; a control unit configured to control operation of the refrigerating appliance; at least one fan, configured to promote heat exchange between said at least one evaporator and the at least one storage compartment, said at least one fan being further configured to be switched between a first operative condition in which said at least one fan is commanded to rotate, and a second operative condition in which said at least one fan is commanded to not rotate. a pressure sensor configured to measure the pressure inside said at least one storage compartment and in signal communication with the control unit for providing a pressure signal proportional to the measure of the pressure inside said at least one storage compartment.
(9) According to an embodiment of the present invention, said pressure sensor is a Micro Electro Mechanical System (MEMS) pressure sensor configured to measure the pressure inside said at least one storage compartment and to transmit to the control unit a corresponding pressure signal proportional to said measured pressure.
(10) By MEMS pressure sensor is herein intended a pressure sensor made up of miniaturized electro-mechanical elements (e.g., between 1 and 100 micrometers in size) manufactured using modified semiconductor device fabrication technologies.
(11) Using a MEMS pressure sensor guarantees several advantages, such as, among others, a highly

precise output capable of efficiently tracking fast pressure variations, and a limited encumbrance.
(12) According to an embodiment of the present invention, the control unit is configured to control the operation of the refrigerating appliance in function of said pressure signal received when said at least one fan is in the first operative condition.
(13) In this way, the control unit is configured to control the operation of the refrigerating appliance by advantageously taking into consideration the way the activation of the at least one fan influences the pressure inside the storage compartment(s).
(14) According to an embodiment of the present invention, said pressure sensor is an absolute pressure sensor.
(15) In this way, assembling the refrigerating appliance is easy, since (unlike the pressure sensors of the differential type) an absolute pressure sensor does not require to be in fluid communication with two different environments, and there is no constraint about the position in the refrigerating appliance where installing the pressure sensor.
(16) According to an embodiment of the present invention, the at least one fan is in signal communication with the control unit, preferably the control unit being configured to set a rotation speed of the at least one fan in function of said pressure signal.
(17) According to an embodiment of the present invention, the control unit is configured to estimate an air flow rate of said at least one fan in function of said pressure signal.
(18) In this way, it is possible to advantageously set the rotation speed of the at least one fan depending on the actual air flow rate of the fan, which causes the pressure of the air of the space where the fan is located.
(19) According to an embodiment of the present invention, the control unit is configured to control said at least one fan so as to: increase the rotation speed of the at least on fan if the pressure signal is below a pressure threshold value; decrease the rotation speed of the at least on fan if the pressure signal is above said pressure threshold value.
(20) In this way, the operation of the refrigerating appliance is improved, since the at least one fan is operating at its optimal rotation speed/air flow rate.
(21) Moreover, it is advantageously possible to dynamically regulate the cooling effect of the at least one evaporator based on the actual density of the air, and therefore based on the actual air flow rate of the at least one fan.
(22) According to an embodiment of the present invention, the control unit is configured to increase the rotation speed of the at least one fan if the pressure signal is indicative of a decrease in the pressure inside the at least one storage compartment with respect to said pressure threshold value.
(23) According to an embodiment of the present invention, the control unit is configured to decrease the rotation speed of the at least one fan if the pressure signal is indicative of an increase in the pressure inside the at least one storage compartment with respect to said pressure threshold value.
(24) According to an embodiment of the present invention, the refrigeration circuit further comprises a compressor for causing refrigerant to flow in the refrigeration circuit through the at least one evaporator, the control unit being in signal communication with the compressor and configured to control the refrigerant flow rate of the compressor in function of said pressure signal.
(25) According to an embodiment of the present invention, the control unit is configured to control the refrigerant flow rate of the compressor by adjusting a duty cycle or a speed of the compressor in function of said pressure signal.
(26) According to an embodiment of the present invention, the refrigerating appliance further comprises a temperature sensor configured to measure the temperature inside said at least one storage compartment and provide a corresponding measured temperature value proportional to the measure of the temperature inside said at least one storage compartment, the control unit being further configured to control the refrigerant flow rate of the compressor in function of said pressure signal and in function of the measured temperature value.

(27) In this way, the refrigerant circuit is advantageously controlled and regulated by taking into consideration not only the temperature of the storage compartment(s), but also the sensed actual air flow rate of the fan. This allow to optimize the power consumption by a corresponding improved control of the compressor operation.
(28) According to an embodiment of the present invention, the control unit is configured to control the refrigerant flow rate of the compressor in function of: said pressure signal, and a comparison between the measured temperature value and a target value.
(29) According to an embodiment of the present invention, the control unit is configured to increase the refrigerant flow rate of the compressor if both the two following conditions are verified: the measured temperature value is higher than a target value, and the estimated air flow rate of the at least one fan is equal to a target air flow rate.
(30) According to an embodiment of the present invention, the target air flow rate is a single value.
(31) According to another embodiment of the present invention, the target air flow rate comprises a range of different values.
(32) According to an embodiment of the present invention, the control unit is configured to decrease the refrigerant flow rate of the compressor and increase the rotation speed of the at least one fan if both the two following conditions are verified: the estimated air flow rate of the at least one fan is lower than a target air flow rate, and the measured temperature value is equal to a target value.
(33) According to an embodiment of the present invention, the control unit is configured to decrease the refrigerant flow rate of the compressor if both the two following conditions are verified: the measured temperature value is lower than a target value, and the estimated air flow rate of the at least one fan is equal to a target flow rate.
(34) According to an embodiment of the present invention, the refrigerating appliance further comprises at least one defrost heater, the control unit being in signal communication with the defrost heater and configured to control the defrost heater in function of said pressure signal.
(35) According to an embodiment of the present invention, the control unit is configured to activate the at least one defrost heater for a corresponding period of time, the control unit being configured to: increase the duration of said corresponding period of time if the pressure signal is indicative of a decrease in the pressure inside said at least one storage compartment; decrease the duration of said corresponding period of time if the pressure signal is indicative of an increase in the pressure inside said at least one storage compartment.
(36) According to an embodiment of the present invention, the control unit is configured to assess fail conditions of the at least one fan in function of said pressure signal received when said at least one fan is in the first operative condition.
(37) In this way, thanks to the precision and the efficient tracking of fast pressure variation of the MEMS pressure sensor, fault conditions of the fan can be detected in a reliable way.
(38) According to an embodiment of the present invention, said at least one fan is configured to be switched between the first operative condition and the second operative condition by the control unit or by a test unit external to the refrigerating appliance.
(39) According to an embodiment of the present invention, said at least one storage compartment comprises a fresh food compartment and a freezer compartment, said pressure sensor being located inside the fresh food compartment or said freezer compartment.
(40) According to an embodiment of the present invention, the refrigerating appliance further comprises: a cabinet enclosing said at least one storage compartment; at least one door configured to provide access to said at least one storage compartment when the at least one door is in an open condition and to prevent access to said at least one storage compartment when the at least one door is in a closed condition; a gasket element mounted along an inner peripheral portion of the at least one door and configured to adhere to a corresponding portion of the cabinet when the at least one door is in the closed condition so as to prevent air in the at least one storage compartment from

leaking out of the at least one storage compartment.
(41) According to an embodiment of the present invention, the control unit is configured to assess gasket element faults causing air leakages in function of said pressure signal received when said at least one fan is in the first operative condition.
(42) In this way, thanks to the precision and the efficient tracking of fast pressure variation of the MEMS pressure sensor, fault conditions of the gasket element can be detected in a reliable way.
(43) According to an embodiment of the present invention, the control unit is configured to assess said gasket element faults if the pressure signal decreases toward a value corresponding to an external ambient pressure.
(44) Another aspect of the present invention relates to a method for operating a refrigerating appliance, the refrigerating appliance comprising: a control unit configured to control operation of the refrigerating appliance at least one storage compartment for storing goods to be refrigerated; a refrigeration circuit comprising at least one evaporator associated with said at least one storage compartment; at least one fan, configured to promote heat exchange between said at least one evaporator and the at least one storage compartment, said at least one fan being further configured to switch between a first operative condition in which said at least one fan is commanded to rotate, and a second operative condition in which said at least one fan is commanded to not rotate; a MEMS pressure sensor configured to measure the pressure inside said at least one storage compartment for providing a pressure signal proportional to a measure of the pressure inside said at least one storage compartment, the method comprising: having the at least one fan in said first operative condition; having the MEMS pressure sensor provide said pressure signal to the control unit; having the control unit control the operation of the refrigerating appliance in function of the pressure signal received from the MEMS pressure when the fan is in the first operative condition.

Description

BRIEF DESCRIPTION OF THE ANNEXED DRAWINGS
(1) These and other features and advantages of the present invention will be made apparent by the following description of some exemplary and non-limitative embodiments thereof; for its better intelligibility, the following description should be read by making reference to the attached drawings, wherein:
(2) FIG. **1** illustrates a refrigerating appliance;
(3) FIG. **2** schematically illustrates a simplified (not-in-scale) cross-sectional side view of the refrigerating appliance of FIG. **1** according to an embodiment of the present invention;
DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS OF THE INVENTION
(4) FIG. **1** illustrates a refrigerating appliance **100** in which concepts according to embodiments of the present invention can be applied.
(5) The refrigerating appliance **100** comprises a number of well known electronic, mechanical and/or electro-mechanical components—however, for the sake of description ease and conciseness, only those being relevant for understanding the invention will be introduced and discussed in the following.
(6) According to an embodiment of the present invention, the refrigerating appliance **100** is a combined-type refrigerating appliance, comprising a first storage compartment **110**(**1**) and a second storage compartment **110**(**2**) where food and beverage articles can be stored and preserved by refrigeration at different temperatures.
(7) In the considered example, the first storage compartment **110**(**1**) is a storage compartment adapted to operate at higher temperature than the second storage compartment **110**(**2**).
(8) For example, the first storage compartment **110**(**1**) is a higher-temperature storage compartment, such as a fresh-food compartment, adapted to be at temperatures above 0° C. (e.g., in

the range [3, 7] ° C.), and the second storage compartment **110**(**2**) is a lower-temperature storage compartment, such as a freezer compartment, adapted to be at temperatures below 0° C. (e.g., in the range [−27, −18] ° C.). These temperature ranges have to be intended only as non limitative examples, since the concepts of the present invention can be directly applied to any range of temperatures, and also if the first and second storage compartments **110**(**1**), **110**(**2**) are both adapted to operate at temperatures below 0° C. or are both adapted to operate at temperatures above 0° C.
(9) In the illustrated example, the first storage compartment **110**(**1**) is positioned above the second storage compartment **110**(**2**), however similar considerations apply in case the positions of the two compartments **110**(**1**), **110**(**2**) are swapped, or in case a single storage compartment only is provided.
(10) Shelves **112** and other structures for supporting and storing food and beverage articles may be provided within both the first and second storage compartments **110**(**1**), **110**(**2**) (in the example illustrated in FIG. **1**, only visible in the first storage compartment **110**(**1**)).
(11) According to an embodiment of the present invention, the refrigerating appliance **100** comprises a substantially parallepiped-shaped cabinet **130** having a top panel **130**(*t*), a rear panel (not visible in FIG. **1**), a bottom panel (not visible in FIG. **1**), side panels **130**(*s*) (only one visible in FIG. **1**). Naturally, similar considerations apply in case the cabinet **130** has a different shape and/or structure.
(12) In order to access the first and second storage compartments **110**(**1**), **110**(**2**), a front section of the cabinet **130** (i.e., the side thereof substantially perpendicular to the top panel **130**(*t*), the bottom panel, and the side panels **130**(*s*)) is provided with respective openings **131**(**1**), **131**(**2**). A first door **135**(**1**) and a second door **135**(**2**) are hingedly mounted to the front section of the cabinet **130** to selectively close/open the openings **131**(**1**) and **131**(**2**), respectively, in order to provide selective access to the first storage compartment **110**(**1**) and to the second storage compartment **110**(**2**), respectively. Different configurations can be also contemplated, such as for example in which a single door is provided for selectively closing/opening the openings **131**(**1**) and **131**(**2**), or in which the refrigerating appliance **100** has a single storage compartment only and a single door.
(13) In FIG. **1**, both the first door **135**(**1**) and a second door **135**(**2**) are depicted in an open condition.
(14) According to a preferred embodiment of the present invention, the first door **135**(**1**) comprises a gasket element **136**(**1**) mounted along an inner peripheral portion of the first door **135**(**1**), and configured to adhere with and remain flush with a corresponding border portion **138**(**1**) of the front section of the cabinet **130** surrounding the opening **131**(**1**) when the first door **135**(**1**) is in the closed condition. In this way, air in the first storage compartment **110**(**1**) is prevented from leaking out of the first storage compartment **110**(**1**) through the opening **131**(**1**) when the first door **135**(**1**) is in the closed condition.
(15) Similarly, according to a preferred embodiment of the present invention, the second door **135**(**2**) comprises a gasket element **136**(**2**) mounted along an inner peripheral portion of the second door **135**(**2**), and configured to adhere with and remain flush with a corresponding border portion **138**(**2**) of the front section of the cabinet **130** surrounding the opening **131**(**2**) when the second door **135**(**2**) is in the closed condition. In this way, air in the second storage compartment **110**(**2**) is prevented from leaking out of the second storage compartment **110**(**2**) through the opening **131**(**2**) when the second door **135**(**2**) is in the closed condition.
(16) Without entering in details well known to those skilled in the art, the gasket elements **136**(**1**) **136**(**2**) are made of a deformable material, such as Polyvinyl chloride (PVC) or neoprene. Preferably, the gasket element **136**(**1**) encloses magnetic material elements, e.g., magnetic strips, which are configured to be attracted toward the border portion **138**(**1**), preferably by magnetic coupling elements **139**(**1**) located at the border portion **138**(**1**). Similarly, the gasket element **136**(**2**) encloses magnetic material elements, e.g., magnetic strips, which are configured to be attracted toward the border portion **138**(**2**), preferably by magnetic coupling elements **139**(**2**) located at the

border portion **138**(**2**).
(17) FIG. **2** schematically illustrates a simplified (not-in-scale) cross-sectional side view of the refrigerating appliance **100** taken from a section plane parallel to the side panels **130**(*s*) according to an embodiment of the present invention.
(18) In FIG. **2**, the rear panel of the cabinet **130** is identified with reference **130**(*r*), and the bottom panel of the cabinet **130** is preferably identified with reference **130**(*b*).
(19) In FIG. **2**, both the first door **135**(**1**) and a second door **135**(**2**) are depicted in a closed condition, with the gasket element **136**(**1**) that is flush with the border portion **138**(**1**) and the gasket element **136**(**2**) that is preferably flush with the border portion **138**(**2**).
(20) According to a preferred embodiment of the present invention, one or more vent holes **137** are provided (only one illustrated in FIG. **2**), each one arranged to form a small and narrow passage through which (small amount of) air can pass between the external environment (i.e., external with respect to the refrigerating appliance **100**) and the storage compartments **110**(**1**), **110**(**2**). The vent holes **137** are expediently used to (relatively slowly) bring back the pressure inside the storage compartments **110**(**1**), **110**(**2**) to the external ambient air pressure following pressure variations inside the storage compartments **110**(**1**), **110**(**2**) caused by perturbations (e.g., the opening/closing of one of the doors **135**(**1**), **135**(**2**)).
(21) According to a preferred embodiment of the present invention, the refrigerating appliance **100** is equipped with a refrigeration circuit for circulating a refrigeration fluid (briefly referred to as "refrigerant").
(22) According to an embodiment of the present invention, the refrigeration circuit comprises a compressor unit **140**, a condenser unit **150**, a fluid expansion unit **160**, a first evaporator unit **170**(**1**) associated to the first storage compartment **110**(**1**), and a second evaporator unit **170**(**2**) associated to the second storage compartment **110**(**2**).
(23) According to an embodiment of the present invention, the two evaporator units **170**(**1**) and **170**(**2**) are fluidly connected in series to each other in the refrigeration circuit, with the second evaporator unit **170**(**2**) that is upstream the first evaporator unit **170**(**2**) along the flow direction of the refrigerant in said refrigeration circuit (illustrated in the figures through bold arrows). It is pointed out that the concepts of the present invention can be directly applied to refrigerating appliances having a different kind of refrigeration circuit, such as for example a refrigerating appliance in which each storage compartment has a respective and independent refrigeration circuit or a refrigeration circuit having one evaporator only fluidically communicating with both compartments.
(24) As it is well known to those skilled in the art, the compressor unit **140** carries out the double function of compressing the refrigerant and causing the circulation of refrigerant itself in the refrigeration circuit—so that the refrigerant flows, in sequence, through the condenser unit **150**, the fluid expansion unit **160**, the second evaporator unit **170**(**2**), and the first evaporator unit **170**(**1**), before reaching again the compressor unit **140**.
(25) According to a preferred embodiment of the present invention, the compressor unit **140** is located on a bottom portion of the refrigerating appliance **100**, preferably close to the rear panel **130**(*r*) of the cabinet **130**.
(26) According to a preferred embodiment of the present invention, the condenser unit **150** is provided on the rear portion of the refrigerating appliance **100**, such as at the rear panel **130**(*r*) of the cabinet **130**.
(27) According to an embodiment of the present invention, the first evaporator unit **170**(**1**) is located at the first storage compartment **110**(**1**), for example close to the rear panel **130**(*r*), and the second evaporator unit **170**(**2**) is located at the second storage compartment **110**(**2**), for example close to the rear panel **130**(*r*).
(28) Naturally, the concepts of the present invention can be directly applied in case one or more among the compressor unit **140**, the condenser unit **150**, the fluid expansion unit **160**, the first

evaporator unit **170**(**1**), and the second evaporator unit **170**(**2**) are located in different positions of the refrigerating appliance **100**, provided that the first evaporator unit **170**(**1**) is arranged in such a way to be able of generating refrigerating air in the first storage compartment **110**(**1**) and the second evaporator unit **170**(**2**) is arranged in such a way to be able of generating refrigerating air in the second storage compartment **110**(**2**).
(29) The compressor unit **140** has an input port fluidly coupled with an output port of the first evaporator unit **170**(**1**) for receiving refrigerant—in the vapor phase—at a relatively low temperature and at a relatively low pressure.
(30) The compressor unit **140** has an output port fluidly coupled with an input port of the condenser unit **150**.
(31) The compressor unit **140** is configured to compress the refrigerant received by the first evaporator unit **170**(**1**) so as to increase the pressure and temperature thereof, and to provide the compressed refrigerant to the condenser unit **150**.
(32) The compressed refrigerant, still in the vapor phase, flows through the condenser unit **150**, wherein it condenses to liquid phase by heat exchange with ambient air.
(33) The condenser unit **150** has an output port fluidly coupled with the fluid expansion unit **160** for providing the refrigerant, now in form of a high pressure liquid, to the latter unit.
(34) The fluid expansion unit **160**, for example an expansion valve or a capillary tube, is configured to reduce the pressure, and the temperature, of the refrigerant.
(35) The refrigerant outputted by the fluid expansion unit **160**—which is a low pressure, low temperature fluid in which liquid and vapor phase coexist—is fed to an input port of the second evaporator unit **170**(**2**).
(36) As the refrigerant flows through the second evaporator unit **170**(**2**), part of the liquid fraction of the refrigerant turns from liquid into vapor through evaporation, causing air inside the second storage compartment **110**(**2**) to cool down.
(37) The refrigerant exits the second evaporator unit **170**(**2**) through an output port thereof, that is fluidly coupled with an input port of the first evaporator unit **170**(**1**).
(38) As the refrigerant flows through the first evaporator unit **170**(**1**), the remaining liquid fraction of the refrigerant turns from liquid into vapor through evaporation, causing air inside the first storage compartment **110**(**1**) to cool down. Once the liquid fraction of the refrigerant is entirely turned into vapor, the temperature of the refrigerant starts to rise up.
(39) According to an embodiment of the present invention, the second evaporator unit **170**(**1**) is smaller than the second evaporator unit **170**(**2**), so that the temperature in the first storage compartment **110**(**1**) is higher than the temperature in the second storage compartment **110**(**2**).
(40) The refrigerant—now in vapor phase—outputted from the output port of the second evaporator unit **170**(**2**) is then fed again to the compressor unit **140**.
(41) According to an embodiment of the present invention, each of or one between the first evaporator unit **170**(**1**) and the second evaporator unit **170**(**2**) may be associated with a respective fan **171**(**1**), **171**(**2**) configured to promote heat exchange between the evaporator itself and the storage compartment associated to the evaporator.
(42) According to an embodiment of the present invention, the compressor unit **140** is capable of setting the volume of refrigerant delivered per time unit (also referred to as "refrigerant flow rate" or "refrigerant throughput"), and therefore setting the temperature inside the storage compartments **110**(**1**) and **110**(**2**) under the control of a control unit **172** configured to control the operation of the refrigerating appliance **100**. For example, the control unit **172** may be configured to generate and send to the compressor unit **140** a compressor driving signal CDS indicative of a desired refrigerant flow rate.
(43) As it is well known to those skilled in the art: if the flow rate of the compressor unit **140** is increased, the cooling effect of the first and second evaporator units **170**(**1**), **170**(**2**) is increased, causing a reduction of the temperatures of the corresponding storage compartments **110**(**1**), **110**(**2**)

if other operative parameters are held constant; if the flow rate of the compressor unit **140** is decreased, the cooling effect of the first and second evaporator units **170**(**1**), **170**(**2**) is reduced, causing an increasing of the temperatures of the corresponding storage compartments **110**(**1**), **110**(**2**) if other operative parameters are held constant.

(44) According to an embodiment of the present invention, the control unit **172** is configured to set the flow rate of the compressor unit **140** by setting the speed thereof, for example to a value proportional to the compressor driving signal CDS.

(45) According to an embodiment of the present invention, the control unit **172** is configured to set the flow rate of the compressor unit **140** by setting the duty cycle thereof, for example to a value proportional to the compressor driving signal CDS.

(46) As it is well known to those skilled in the art, the higher the speed or the duty cycle of the compressor unit **140**, the higher the flow rate. Conversely, the lower the speed or the duty cycle of the compressor unit **140**, the lower the flow rate.

(47) According to an embodiment of the present invention, the refrigerating appliance **100** comprises a user interface UI through which a user may input commands to the control unit **172**. For example, the user interface UI can be advantageously exploited for setting a desired temperature for the first storage compartment **110**(**1**), hereinafter referred to as first target temperature Tt(**1**), and a desired temperature for the second storage compartment **110**(**2**), hereinafter referred to as second target temperature Tt(**2**).

(48) Different kind of user interfaces UI can be provided, from simple rotary knobs inside the storage compartments, to more advanced user interfaces provided with touch screen, buttons, and/or visual display located on one or more of the doors and/or on cabinet. Still more advanced user interfaces UI can be considered, comprising a wireless communication unit adapted to receive commands and provide data from/to a personal user device, such as a smartphone.

(49) According to an embodiment of the present invention, temperature sensors are provided in both the first storage compartment **110**(**1**) and the second storage compartment **110**(**2**) for sensing the temperature thereinside.

(50) Particularly, according to a preferred embodiment of the present invention, a first temperature sensor **175**(**1**) is located inside the first storage compartment **110**(**1**) and a second temperature sensor **175**(**2**) is located inside the second storage compartment **110**(**2**).

(51) According to a preferred embodiment of the present invention, the first and second temperature sensors **175**(**1**), **175**(**2**) are connected to the control unit **172**, to provide the sensed temperatures to the latter. The first temperature sensor **175**(**1**) is configured to provide a first sensed temperature Ts(**1**) value indicative of the temperature inside the first storage compartment **110**(**1**) to the control unit **172**, and the second temperature sensor **175**(**2**) is configured to provide a second sensed temperature Ts(**2**) value indicative of the temperature inside the second storage compartment **110**(**2**) to the control unit **172**.

(52) According to an embodiment of the present invention, the control unit **172** is configured to set the flow rate of the compressor unit **140** in function of the first sensed temperature Ts(**1**), to the second sensed temperature Ts(**2**), to the first target temperature Tt(**1**), and to the second target temperature Tt(**2**). For example, the control unit **172** may be configured to set the flow rate of the compressor unit **140** in function of a first difference D(**1**)=Ts(**1**)−Tt(**1**) between the first sensed temperature Ts(**1**) and the first target temperature Tt(**1**), and in function of a second difference D(**2**)=Ts(**2**)−Tt(**2**) between the second sensed temperature Ts(**2**) and the second target temperature Tt(**2**). For example, the control unit **172** may be configured to set the flow rate of the compressor unit **140** in function of a value corresponding to an average of the first and second differences D(**1**), D(**2**).

(53) According to an embodiment of the present invention, each of the first fan **171**(**1**) and the second fan **17**(**2**) is configured to be switched between a first operative condition (hereinafter, “active operative condition”) in which the fan is commanded to rotate, and a second operative

condition (hereinafter, “inactive operative condition”) in which the fan is commanded to not rotate.
(54) For example, according to an embodiment of the present invention, the control unit **172** is configured to switch the first fan **171**(**1**) to the active operative condition by generating and sending to the first fan **171**(**1**) a corresponding first fan driving signal FS(**1**). Similarly, according to an embodiment of the present invention, the control unit **172** is configured to switch the second fan **171**(**2**) to the active operative condition by generating and sending to the second fan **171**(**2**) a corresponding second fan driving signal FS(**2**).
(55) According to a preferred embodiment of the present invention, the control unit **172** is configured to set the rotation speed of the first fan **171**(**1**) and of the second fan **171**(**2**).
(56) For example, the control unit **172** may be configured to set the rotation speed of the first fan **171**(**1**) to a value depending on (e.g., proportional to) the first fan driving signal FS(**1**). Similarly, the control unit **172** may be configured to set the rotation speed of the second fan **171**(**2**) to a value depending on (e.g., proportional to) the second fan driving signal FS(**2**).
(57) As it is known to those skilled in the art, if other operative parameters are held constant, the volume of air moved per time unit (also referred to as “air flow rate”) of a fan depends on (i.e., it is proportional to) the rotation speed of the fan itself. Therefore, by setting the rotation speed of the first fan **171**(**1**) it is possible to set also the air flow rate of the first fan **171**(**1**), and by setting the rotation speed of the second fan **171**(**2**) it is possible to set also the air flow rate of the second fan **171**(**2**).
(58) If the flow rate of the first fan **171**(**1**) is increased, the thermal exchange between the first evaporator **170**(**1**) and the volume of air inside the first storage compartment **110**(**1**) is improved, and therefore, if other operative parameters are held constant, the temperature inside the first storage compartment **110**(**1**) is decreased.
(59) If the flow rate of the first fan **171**(**1**) is decreased, the heat exchange between the first evaporator **170**(**1**) and the volume of air inside the first storage compartment **110**(**1**) is reduced, and therefore, if other operative parameters are held constant, the temperature inside the first storage compartment **110**(**1**) is increased.
(60) If the flow rate of the second fan **171**(**2**) is increased, the heat exchange between the second evaporator **170**(**2**) and the volume of air inside the second storage compartment **110**(**2**) is improved, and therefore, if other operative parameters are held constant, the temperature inside the second storage compartment **110**(**2**) is decreased.
(61) If the flow rate of the second fan **171**(**2**) is decreased, the heat exchange between the second evaporator **170**(**2**) and the volume of air inside the second storage compartment **110**(**2**) is reduced, and therefore, if other operative parameters are held constant, the temperature inside the second storage compartment **110**(**2**) is increased.
(62) According to a preferred embodiment of the present invention, the control unit **172** is configured to set the rotation speed (and therefore the air flow rate) of the first fan **171**(**1**) in function of the first difference D(**1**), i.e., in function of the difference between the first sensed temperature Ts(**1**) and the first target temperature Tt(**1**).
(63) According to a preferred embodiment of the present invention, the control unit **172** is configured to set the rotation speed (and therefore the air flow rate) of the second fan **171**(**2**) in function of the second difference D(**2**), i.e., in function of the difference between the second sensed temperature Ts(**2**) and the second target temperature Tt(**2**).
(64) According to an embodiment of the present invention, the refrigerating appliance **100** comprises a pressure sensor **180** located inside one of the storage compartments **110**(**1**), **110**(**2**). In the embodiment of the invention illustrated in FIG. **2**, the pressure sensor **180** is located in the first storage compartment **110**(**2**), however, similar considerations apply if the pressure sensor **180** is located in the second storage compartment. Solutions are contemplated in which more than one pressure sensors **180** are provided, for example each one located in a respective storage compartment.

(65) According to an embodiment of the present invention, the pressure sensor **180** is in signal communication with the control unit **172**.
(66) According to an embodiment of the present invention, the pressure sensor **180** is configured to output a corresponding pressure signal PS proportional to the measured pressure to the control unit **172**.
(67) According to an embodiment of the present invention, said pressure signal PS is a digital signal.
(68) According to a preferred embodiment of the present invention, the pressure sensor **180** is a Micro Electro Mechanical System (MEMS) pressure sensor configured to measure the pressure inside the storage compartment **110**(**1**), **110**(**2**) wherein it is located.
(69) According to an embodiment of the present invention, the pressure sensor **180** is a MEMS capacitive pressure sensor, comprising a first conductive layer deposed on a flexible diaphragm suspended over a cavity, and a second conductive layer deposed on the bottom of the cavity. In this way, a capacitor is formed, the electric capacitance thereof changing in response to deflections of the diaphragm caused by the pressure.
(70) According to another embodiment of the present invention, the pressure sensor **180** is a MEMS piezoresistive pressure sensor, comprising piezoresistive conductive layers deposed on a suspended flexible diaphragm. The electric resistance of the piezoresistive conductive layers change in response to deflections of the diaphragm caused by the pressure.
(71) According to an embodiment of the present invention, the (MEMS) pressure sensor **180** is an absolute pressure sensor, i.e., a pressure sensor having a single port, which is in fluid communication with the environment the pressure thereof is intended to be measured. In the exemplary embodiment of the invention illustrated in FIG. **2**, this environment in fluid communication with said single port of the pressure sensor **180** is the first storage compartment **110**(*i*).
(72) Thanks to the intrinsic simplicity (from the interconnection point of view) of the pressure sensor **180** according to the embodiment of the present invention, the feasible locations for installing the pressure sensor **180** in the refrigerating appliance **100** are many, because an absolute pressure sensor does not require the provision of tube elements.
(73) For example, according to an advantageous embodiment of the present invention, the pressure sensor **180** is advantageously mounted on the circuit board **176** where the control unit **172** is located. In this way, no complicated wiring is required for sending the pressure signal PS generated by the pressure sensor **180** to the control unit **172**, since both the former and the latter are located on a same circuit board. Moreover, the pressure sensor **180** may by advantageously powered exploiting the same power supply used by the other components located on the circuit board **176**, like the same power supply used for supplying the control unit **172**.
(74) Naturally, similar considerations apply in case the pressure sensor **180** is located in a different portion of the storage compartment **110**(**1**) (or **110**(**2**)). For example, the pressure sensor **180** can be installed in the same circuit board wherein Light-Emitting Diodes (LEDs) (not illustrated in the figures) for the illumination of the storage compartment **110**(**1**) (or **110**(**2**)) are provided.
(75) According to an embodiment of the present invention, the control unit **172** is configured to control the operation of the refrigerating appliance **100** based on the pressure signal PS generated by the pressure sensor **180** when at least one between the first and second fan **171**(**1**), **171**(**2**) are in the active operative condition, i.e., when at least one between the first and second fan **171**(**1**), **171**(**2**) is commanded to rotate.
(76) In other words, according to an embodiment of the present invention, the control unit **172** is configured to control the operation of the refrigerating appliance **100** by advantageously taking into consideration the way the activation of the first and second fan **171**(**1**), **171**(**2**) influence the pressure inside the storage compartment **110**(**1**) and/or **1102**(**2**).
(77) According to an embodiment of the present invention, the information about the pressure

inside the storage compartment **110**(**1**) and/or **1102**(**2**) is advantageously exploited by the control unit **172** to properly set the rotation speed of the first fan **171**(**1**) and/or the second fan **171**(**2**). More in particular, since the air flow rate of a fan depends on the pressure of the air of the space wherein the fan is located, by knowing the pressure inside the storage compartment **110**(**1**) and/or the storage compartment **110**(**2**) when the first fan **110**(**1**) and/or second fan **110**(**2**) is/are commanded to rotate it is possible to estimate the air flow rate of the first fan **110**(**1**) and/or second fan **110**(**2**).
(78) Indeed, given a rotation speed value of a fan, the higher the pressure of the storage compartment where the fan is located, the higher the air flow rate of said fan because of a higher air density. Similarly, given a rotation speed value of a fan, the lower the pressure of the storage compartment where the fan is located, the lower the air flow rate of said fan because of a lower air density.
(79) In view of the above, and by making reference to the exemplary embodiment of the present invention illustrated in FIG. **2** in which the pressure sensor **180** is located in the first storage compartment **110**(**1**), according to an embodiment of the present invention the control unit **172** is configured to set the rotation speed of the first fan **171**(**1**) in function of the pressure signal PS generated by the pressure sensor **180**.
(80) According to an embodiment of the present invention, the control unit **172** is configured to estimate the air flow rate of the first fan **171**(**1**) according to the pressure signal PS.
(81) According to an embodiment of the present invention, the control unit **172** is configured to control the first fan **171**(**1**) in such a way to: increase the rotation speed of the first fan **171**(**1**) if the pressure signal PS has a value lower than a corresponding pressure threshold value PTH; decrease the rotation speed of the first fan **171**(**1**) if the pressure signal PS has a value higher than the pressure threshold value PTH.
(82) Indeed, a low pressure (lower than the pressure threshold value PTH) may cause a too low air flow rate of the fan, which can be compensated by increasing the rotation speed of the fan.
(83) Similarly, a high pressure (higher than the pressure threshold value PTH) may cause an increase of the air flow rate of the fan that can be advantageously exploited to reduce the rotation speed of the fan for obtaining a desired target air flow rate with a reduced power consumptions.
(84) Naturally, similar considerations apply in case the pressure sensor **180** is located in the second storage compartment **110**(**2**). In this case, according to an embodiment of the invention, the control unit **172** is configured to estimate the air flow rate of the second fan **171**(**2**) in function of the pressure signal PS, and/or according to an embodiment of the present invention, the control unit **172** is configured to control the second fan **171**(**2**) in such a way to: increase the rotation speed of the second fan **171**(**2**) (e.g., by increasing the value of the first fan driving signal FS(**1**)) if the pressure signal PS has a value lower than a corresponding pressure threshold value PTH; decrease the rotation speed of the second fan **171**(**2**) (e.g., by decreasing the value of the first fan driving signal FS(**1**)) if the pressure signal PS has a value higher than the pressure threshold value PTH.
(85) According to an embodiment of the present invention, the control unit **172** is configured to increase the rotation speed of the first fan **171**(**1**) (or, in case the pressure sensor **180** is in the second storage compartment **110**(**2**), of the second fan **171**(**2**)) if the pressure signal PS is indicative of a decrease in the pressure inside the first storage compartment **110**(**1**) with respect to the pressure threshold value PTH.
(86) According to an embodiment of the present invention, the control unit **172** is configured to decrease the rotation speed of the first fan **171**(**1**) (or, in case the pressure sensor **180** is in the second storage compartment **110**(**2**), of the second fan **171**(**2**)) if the pressure signal PS is indicative of an increase in the pressure inside the first storage compartment **110**(**1**) with respect to the pressure threshold value PTH.
(87) According to an embodiment of the present invention, the control unit **172** is configured to control the refrigerant flow rate of the compressor unit **140** in function of the pressure signal PS

generated by the pressure sensor **180**, in such a way to dynamically regulate the cooling effect of the first and second evaporator units **170**(**1**), **170**(**2**) based on the actual density of the air, and therefore based on the actual air flow rate of the first and/or second fan **171**(**1**), **171**(**2**).

(88) By making reference to the exemplary embodiment of the invention illustrated in FIG. **2**, in which the pressure sensor **180** is located in the storage compartment **110**(**1**), since the higher the pressure signal PS, the higher the air flow rate of the first fan **171**(**1**) (if other operative parameters are held constant), according to an embodiment of the present invention the control unit **172** may advantageously decrease the cooling effect of the first and second evaporator units **170**(**1**), **170**(**2**) by decreasing the refrigerant flow rate (e.g., by setting the speed or the duty cycle of the compressor unit **140** through the compressor driving signal CDS) in case the pressure signal PS has a high value. Indeed, in this particular situation, the refrigerant flow rate of the compressor unit **140** may be advantageously reduced, reducing thus electric power consumption, without impairing the overall efficacy of the refrigerating appliance **100**, since the reduced cooling effect of the first evaporator unit **170**(**1**) is compensated by an increased heat exchange between the first evaporator unit **170**(**1**) and the storage compartment **110**(**1**) promoted by the first fan **171**(**1**) in a high air pressure condition.

(89) Similarly, since the lower the pressure signal PS, the lower the air flow rate of the first fan **171**(**1**) (if other operative parameters are held constant), according to an embodiment of the present invention the control unit **172** may advantageously increase the cooling effect of the first and second evaporator units **170**(**1**), **170**(**2**) by increasing the refrigerant flow rate of the compressor unit **140** (e.g., by setting the speed or the duty cycle thereof through the compressor driving signal CDS) in case the pressure signal PS has a low value. Indeed, in this particular situation, the refrigerant flow rate of the compressor unit **140** may be advantageously increased to increase the cooling effect of the first evaporator unit **170**(**1**) to compensate for the reduced heat exchange between the first evaporator unit **170**(**1**) and the storage compartment **110**(**1**) promoted by the first fan **171**(**1**) in a low air pressure condition.

(90) Similar considerations apply in case the pressure sensor **180** is located in the second storage compartment **110**(**2**).

(91) According to an embodiment of the present invention, the control unit **172** is configured to control the refrigerant flow rate of the compressor unit **140** (e.g., by setting the speed or the duty cycle thereof through the compressor driving signal CDS) in function of the pressure signal PS generated by the pressure sensor **180** and the temperature sensed by the temperature sensor located in the storage compartment wherein the pressure sensor **180** is located.

(92) By making reference to the exemplary embodiment of the invention illustrated in FIG. **2**, wherein the pressure sensor **180** is located in the storage compartment **110**(**1**), the control unit **172** is configured to control the refrigerant flow rate of the compressor unit **140** in function of the pressure signal PS generated by the pressure sensor **180** and in function of the first sensed temperature Ts(**1**) measured by the first temperature sensor **175**(**1**).

(93) According to an embodiment of the present invention, the control unit **172** is configured to control the refrigerant flow rate of the compressor unit **140** in function of the pressure signal PS and a comparison between the first sensed temperature Ts(**1**) and the first target temperature Tt(**1**).

(94) More in detail, according to an embodiment of the present invention, the control unit **172** is configured to estimate the air flow rate of the first fan **171**(**1**) in function of the pressure signal PS, and increase the refrigerant flow rate of the compressor unit **140** if both the two following conditions are verified: the first sensed temperature Ts(**1**) is higher than the first target temperature Tt(**1**), and the estimated air flow rate of the first fan **171**(**1**) is equal to or higher than a target air flow rate TFR(**1**).

(95) In this situation, the control unit **172** is aware that the temperature in the first storage compartment **110**(**1**) is too high, while the actual air flow rate of the first fan **171**(**1**) is already at the desired target air flow rate TFR(**1**). Therefore, the best way for reducing the temperature in the

first storage compartment **110**(**1**) without causing an undesired variation of the actual air flow rate of the first fan **171**(**1**) is to increase the cooling effect of the first evaporator unit **170**(**1**) by increasing the refrigerant flow rate of the compressor unit **140**.
(96) According to an embodiment of the present invention, the target air flow rate TFR(**1**) is a single value.
(97) According to another embodiment of the present invention, the target air flow rate TFR(**1**) may comprises a range of different values.
(98) According to an embodiment of the present invention, the control unit **172** is configured to estimate the air flow rate of the first fan **171**(**1**) in function of the pressure signal PS, and decrease the refrigerant flow rate of the compressor unit **140** if both the two following conditions are verified: the first sensed temperature Ts(**1**) is lower than the first target temperature Tt(**1**), and the estimated air flow rate of the first fan **171**(**1**) is equal to or higher than the target air flow rate TFR(**1**).
(99) In this situation, the control unit **172** is aware that the temperature in the first storage compartment **110**(**1**) is too low, while the actual air flow rate of the first fan **171**(**1**) is already at the desired target air flow rate TFR(**1**). Therefore, the best way for increasing the temperature in the first storage compartment **110**(**1**) without causing an undesired variation of the actual air flow rate of the first fan **171**(**1**) is to decrease the cooling effect of the first evaporator unit **170**(**1**) by decreasing the refrigerant flow rate of the compressor unit **140**.
(100) According to an embodiment of the present invention, the control unit **172** is configured to estimate the air flow rate of the first fan **171**(**1**) in function of the pressure signal PS, and decrease the refrigerant flow rate of the compressor unit **140** and at the same time increase the rotation speed of the first fan **171**(**1**) if both the two following conditions are verified: the estimated air flow rate of the first fan **171**(**1**) is lower than the target air flow rate TFR(**1**), and the first sensed temperature Ts(**1**) is equal to the first target temperature Tt(**1**).
(101) In this situation, the control unit **172** is aware that while the temperature in the first storage compartment **110**(**1**) is at the desired value TT(**1**), the actual air flow rate of the first fan **171**(**1**) is lower than the desired target air flow rate TFR(**1**). Therefore, the rotation speed of the first fan **171**(**1**) is increased to bring the air flow rate of the first fan **171**(**1**) to the target air flow rate TFR(**1**), and the potential and undesired temperature decrease caused by such increased rotation speed of the first fan **171**(**1**) is counterbalanced by a reduced refrigerant flow rate of the compressor unit **140**.
(102) Naturally, similar considerations apply in case the pressure sensor **180** is located in the second storage compartment **110**(**2**), with the control unit **172** that is configured to take into account the second sensed temperature Ts(**2**), the second target temperature Tt(**2**), and a target air flow rate TFR(**2**) of the second fan **171**(**2**).
(103) According to an embodiment of the present invention, the pressure inside the storage compartment **110**(**1**) and/or **110**(**2**) measured by the pressure sensor **180** may be also exploited to assess condensation/dew point conditions of the refrigerating appliance **100** during the operation thereof.
(104) According to an embodiment of the present invention, the refrigerating appliance **100** further comprises one or more defrost heaters (only one depicted in FIG. **2**, and identified with reference **198**) adapted to be driven by the control unit **172** to be (e.g., periodically) activated for melting frost formed on internal walls of the storage compartments because of dew. According to an embodiment of the present invention, the control unit **172** is configured to drive the defrost heater **198** by transmitting a defrost activation command DF to the latter for the activation thereof.
(105) According to an embodiment of the present invention, the control unit **172** is configured to activate the defrost heater **198** (through the defrost activation command DF) and keep activated said defrost heater **198** for a corresponding time period (defrost period DP).
(106) According to an embodiment of the present invention, the control unit **172** is configured to

control the defrost heater **195** in function of the pressure signal PS generated by the pressure sensor **180**.

(107) According to an embodiment of the present invention, the control unit **172** is configured to increase the defrost period DP during which the defrost heater **198** is kept activated if the pressure signal PS is indicative of a decrease of pressure inside the first storage compartment **110**(**1**). Indeed, in this condition, the dew point temperature is decreased, and a longer defrost period DP is required.

(108) Similarly, the control unit **172** is configured to decrease the defrost period DP during which the defrost heater **198** is kept activated if the pressure signal PS is indicative of an increase of pressure inside the first storage compartment **110**(**1**). Indeed, in this condition, the dew point temperature is increased, and a shorter defrost period DP is required.

(109) According to an embodiments of the present invention, the precise and reliable pressure measure that can be obtained thanks to the pressure sensor **180** is advantageously exploited to assess fail conditions of the fan **171**(**1**) and/or **171**(**2**).

(110) By making reference to the exemplary embodiment of the invention illustrated in FIG. **2**, wherein the pressure sensor **180** is located in the storage compartment **110**(**1**), the control unit **172** is configured to assess a fail condition of the fan **171**(**1**) in function of the pressure signal PS when the fan **171**(**1**) is in the active operative condition, i.e., when it is commanded to rotate.

(111) When the fan **171**(**1**) is in the active operative condition (and therefore it is in a condition in which it is commanded to rotate) and actually rotates, a pressure variation is caused inside the storage compartment **110**(**1**) that is detected by the pressure sensor **180**. In this situation, the control unit **172** assesses that the fan **171**(**1**) is operating in a correct way.

(112) When the fan **171**(**1**) is in the active operative condition (and therefore it is in a condition in which it is commanded to rotate) but does not rotate, no pressure variation is caused inside the storage compartment **110**(**1**). In this situation, the control unit **172** assesses that the fan **171**(**1**) is affected by a fault.

(113) This procedure can be carried out by the control unit **172** periodically, so as to implement an automatic diagnostic function.

(114) According to another embodiment of the present invention, when the fan **171**(**1**) is in the active operative condition and it is commanded to rotate to generate an air flow rate corresponding to the target air flow rate TFR(**1**), but the fan **171**(**1**) is not capable of reaching the target air flow rate TFR(**1**), this fan condition can be advantageously assessed by the control unit **172** by receiving from the pressure sensor **180** a pressure signal PS indicative of a too low pressure inside the storage compartment **110**(**1**).

(115) Naturally, similar considerations apply to the second fan **171**(**2**) in case the pressure sensor **180** is located in the second storage compartment **110**(**2**).

(116) As already described in the previous, according to an embodiment of the present invention, the switching of the fans **171**(**1**), **171**(**2**) between the active operative condition and the inactive operative condition is carried out by the control unit **172** through the generation and sending of the first fan driving signal FS(**1**) and of the second fan driving signal FS(**2**). However, the concepts of the present invention can be applied also in case the switching of the fans **171**(**1**), **171**(**2**) between the active operative condition and the inactive operative condition during test operations is carried out by a proper test unit external to the refrigerating appliance **100**.

(117) According to an embodiments of the present invention, the precise and reliable pressure measure that can be obtained thanks to the pressure sensor **180** is advantageously exploited to assess faults in the gasket element(s) **136**(**1**) and/or **136**(**2**) causing air leakages.

(118) By making reference to the exemplary embodiment of the invention illustrated in FIG. **2**, wherein the pressure sensor **180** is located in the storage compartment **110**(**1**), the control unit **172** is configured to assess a fault in the gasket element **136**(**1**) causing air leakage in function of the pressure signal PS when the fan **171**(**1**) is in the active operative condition, i.e., when it is

commanded to rotate.
(119) According to an embodiment of the present invention, the control unit **172** is configured to assess a fault in the gasket element **136**(**1**) causing air leakage if the pressure signal PS measured by the pressure sensor **180** when the first an **171**(**1**) is in the active operative condition decreases toward a value corresponding to an external ambient temperature. Indeed, the pressure increase caused by the rotation of the first fan **171**(**1**) is compensated by the air leakage caused by the fault in the gasket element **136**(**1**). In other words, if the gasket element **136**(**1**) has a fault causing air leakage, the pressure inside the first storage compartment **110**(**1**) is brought toward the external ambient pressure irrespective of the rotation of the fan **171**(**1**).
(120) Similar considerations apply to the second fan **171**(**2**) in case the pressure sensor **180** is located in the second storage compartment **110**(**2**).
(121) Naturally, in order to satisfy local and specific requirements, a person skilled in the art may apply to the invention described above many logical and/or physical modifications and alterations. More specifically, although the invention has been described with a certain degree of particularity with reference to preferred embodiments thereof, it should be understood that various omissions, substitutions and changes in the form and details as well as other embodiments are possible. In particular, different embodiments of the invention may even be practiced without the specific details (such as the numeric examples) set forth in the preceding description for providing a more thorough understanding thereof; on the contrary, well known features may have been omitted or simplified in order not to obscure the description with unnecessary particulars.
(122) For example, although in the embodiments of the invention herein described the refrigerating appliance comprises two storage compartments, the concepts of the present invention can be directly applied in case more than two storage compartments are provided.

Claims

1. A refrigerating appliance comprising: at least one storage compartment for storing goods to be refrigerated; a cabinet enclosing said at least one storage compartment; at least one door configured to provide access to said at least one storage compartment when the at least one door is in an open condition and to prevent access to said at least one storage compartment when the at least one door is in a closed condition; a gasket element mounted along an inner peripheral portion of the at least one door and configured to adhere to a corresponding portion of the cabinet when the at least one door is in the closed condition so as to prevent air in the at least one storage compartment from leaking out of the at least one storage compartment; a refrigeration circuit comprising at least one evaporator associated with said at least one storage compartment; a control unit configured to control operation of the refrigerating appliance; at least one fan, configured to promote heat exchange between said at least one evaporator and the at least one storage compartment, said at least one fan being further configured to be switched between a first operative condition in which said at least one fan is commanded to rotate, and a second operative condition in which said at least one fan is commanded to not rotate; and a MEMS pressure sensor configured to measure the pressure inside said at least one storage compartment and in signal communication with the control unit for providing a pressure signal proportional to the measure of the pressure inside said at least one storage compartment, wherein the control unit is configured to control the operation of the refrigerating appliance in function of said pressure signal received when said at least one fan is in the first operative condition, and wherein the control unit is configured to assess gasket element faults causing air leakages in function of said pressure signal received when said at least one fan is in the first operative condition.
2. The refrigerating appliance of claim 1, wherein said pressure sensor is an absolute pressure sensor.
3. The refrigerating appliance of claim 1, wherein the at least one fan is in signal communication

with the control unit, preferably the control unit being configured to set a rotation speed of the at least one fan in function of said pressure signal.

4. The refrigerating appliance of claim 3, wherein the control unit is configured to estimate an air flow rate of said at least one fan in function of said pressure signal.

5. The refrigerating appliance of claim 3, wherein the control unit is configured to control said at least one fan so as to: increase the rotation speed of the at least on fan if the pressure signal is below a pressure threshold value; decrease the rotation speed of the at least on fan if the pressure signal is above said pressure threshold value.

6. The refrigerating appliance of claim 1, wherein the refrigeration circuit further comprises a compressor for causing refrigerant to flow in the refrigeration circuit through the at least one evaporator, the control unit being in signal communication with the compressor and configured to control the refrigerant flow rate of the compressor in function of said pressure signal.

7. The refrigerating appliance of claim 6, wherein the control unit is configured to control the refrigerant flow rate of the compressor by adjusting a duty cycle or a speed of the compressor in function of said pressure signal.

8. The refrigerating appliance of claim 6, further comprising a temperature sensor configured to measure the temperature inside said at least one storage compartment and provide a corresponding measured temperature value proportional to the measure of the temperature inside said at least one storage compartment, the control unit being further configured to control the refrigerant flow rate of the compressor in function of said pressure signal and in function of the measured temperature value.

9. The refrigerating appliance of claim 8, wherein the control unit is configured to control the refrigerant flow rate of the compressor in function of: said pressure signal, and a comparison between the measured temperature value and a target value.

10. The refrigerating appliance of claim 8, wherein the control unit is configured to increase the refrigerant flow rate of the compressor if both the two following conditions are verified: the measured temperature value is higher than a target value, and an estimated air flow rate of the at least one fan is equal to a target air flow rate.

11. The refrigerating appliance of claim 8, wherein the control unit is configured to decrease the refrigerant flow rate of the compressor and increase the rotation speed of the at least one fan if both the two following conditions are verified: an estimated air flow rate of the at least one fan is lower than a target air flow rate, and the measured temperature value is equal to a target value.

12. The refrigerating appliance of claim 8, wherein the control unit is configured to decrease the refrigerant flow rate of the compressor if both the two following conditions are verified: the measured temperature value is lower than a target value, and an estimated air flow rate of the at least one fan is equal to a target flow rate.

13. The refrigerating appliance of claim 1, further comprising at least one defrost heater, the control unit being in signal communication with the defrost heater and configured to control the defrost heater in function of said pressure signal.

14. The refrigerating appliance of claim 13, wherein the control unit is configured to activate the at least one defrost heater for a corresponding period of time, the control unit being configured to: increase the duration of said corresponding period of time if the pressure signal is indicative of a decrease in the pressure inside said at least one storage compartment; decrease the duration of said corresponding period of time if the pressure signal is indicative of an increase in the pressure inside said at least one storage compartment.

15. The refrigerating appliance of claim 1, wherein the control unit is configured to assess fail conditions of the at least one fan in function of said pressure signal received when said at least one fan is in the first operative condition.

16. The refrigerating appliance of claim 1, wherein said at least one fan is configured to be switched between the first operative condition and the second operative condition by the control

unit or by a test unit external to the refrigerating appliance.

17. The refrigerating appliance of claim 1, wherein said at least one storage compartment comprises a fresh food compartment and a freezer compartment, said pressure sensor being located inside the fresh food compartment or said freezer compartment.

18. The refrigerating appliance of claim 1, wherein the control unit is configured to assess said gasket element faults if the pressure signal decreases toward a value corresponding to an external ambient pressure.

19. A method for operating a refrigerating appliance, the refrigerating appliance comprising: a control unit configured to control operation of the refrigerating appliance; at least one storage compartment for storing goods to be refrigerated; a cabinet enclosing said at least one storage compartment; at least one door configured to provide access to said at least one storage compartment when the at least one door is in an open condition and to prevent access to said at least one storage compartment when the at least one door is in a closed condition; a gasket element mounted along an inner peripheral portion of the at least one door and configured to adhere to a corresponding portion of the cabinet when the at least one door is in the closed condition so as to prevent air in the at least one storage compartment from leaking out of the at least one storage compartment; a refrigeration circuit comprising at least one evaporator associated with said at least one storage compartment; at least one fan, configured to promote heat exchange between said at least one evaporator and the at least one storage compartment, said at least one fan being further configured to switch between a first operative condition in which said at least one fan is commanded to rotate, and a second operative condition in which said at least one fan is commanded to not rotate; and a MEMS pressure sensor configured to measure the pressure inside said at least one storage compartment for providing a pressure signal proportional to the measure of the pressure inside said at least one storage compartment, the method comprising: having the at least one fan in said first operative condition; having the MEMS pressure sensor provide said pressure signal to the control unit; having the control unit control the operation of the refrigerating appliance in function of the pressure signal received from the MEMS pressure sensor when the at least one fan is in the first operative condition; and having the control unit assess gasket element faults causing air leakages in function of said pressure signal received when said at least one fan is in the first operative condition.